Detection-Driven Underwater Image Enhancement

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01 Problem statement

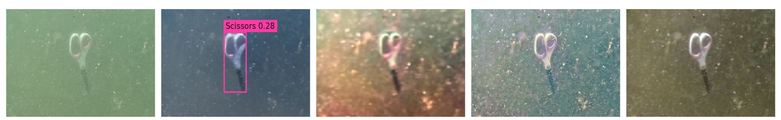
Underwater image enhancement(UIE) can improve perceptual quality of an image but do not improve downstream robot vision tasks like object detection.

Can detection supervision help with robotic perception?

UIE on Lake Images → YOLO26x Detection

mAP50 (%) · same YOLO26x checkpoint, imgsz=640

Original	DUnIE-Diff	Osmosis [2]	WWE-UIE [3]	SLURPP [1]
25.88	**34.16**	4.35	21.45	19.30



Not detected | **Scissors 0.28** | Not detected | Not detected | Not detected

Scissors-only mAP50; enhancement resolution not controlled.

Paired pool training data

SeaThru + Depth Anything V2 synthesize lake-like distortion of paired Pool COU [5] frames.

Original pool reference



Synthetic distorted input



2,858 pairs | 11 classes | 6,006 objects

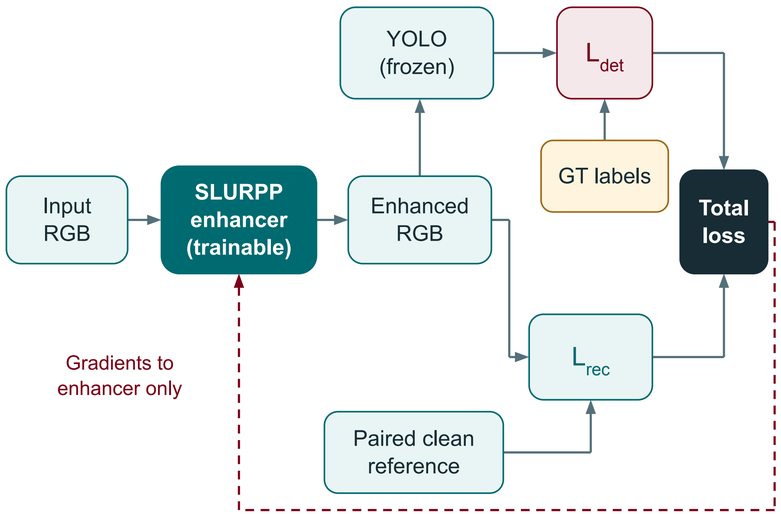
Training	Validation	Test
2,286 pairs	286 pairs	286 pairs

Depth: 0.5–6.5 pseudo-m (not measured range).
Pool originals are restoration references.

02 Methodology

Fine-tune a pretrained SLURPP [1] diffusion enhancer using paired images and object annotations.

TRAINING



$$L = \lambda_{rec} L_{rec} + \lambda_{det} L_{det}$$

Reconstruction uses L1 + 0.1 × VGG perceptual loss. YOLO supplies detection loss. YOLO and VGG stay frozen while gradients update the enhancer.

INFERENCE

Equally weighted L_{det} and L_{rec} give the best results

Resize to 512 × 512 → one prediction step → enhanced RGB

03 Preliminary results

65.29% mAP50

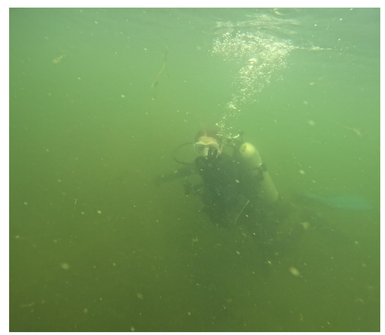
YOLO11m trained on VDD-C [6]
DUnIE-Diff vs WWE-UIE: +9.46 percentage points in mAP50.

Method	mAP50
Original	0.00
DUnIE-Diff	**65.29**
Osmosis [2]	26.18
WWE-UIE [3]	55.83
SLURPP [1]	49.77

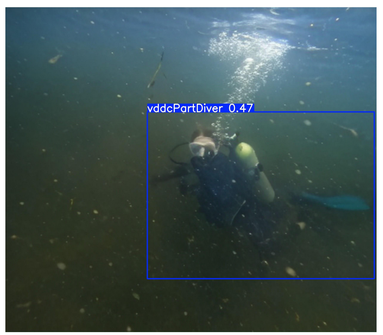
Highest mAP50 on diver frames

DUnIE-Diff achieves the highest mAP50.
Output sizes differ: Osmosis 256×256, SLURPP 512×512, WWE-UIE 3840×3360. Resolution is not controlled.

Original: no detection



DUnIE-Diff: YOLO confidence 0.47



04 Ongoing work

- **Separate data and model effects**
- **Validate generalization**
- **Robotic deployment**

05 References

[1] J. Wu et al., “Single-Step Latent Diffusion for Underwater Image Restoration,” IEEE TPAMI, 2025. (SLURPP)
[2] O. Bar Nathan et al., “Osmosis: RGBD Diffusion Prior for Underwater Image Restoration,” ECCV, 2024.
[3] C.-H. Cheng et al., “WWE-UIE: A Wavelet & White Balance Efficient Network for Underwater Image Enhancement,” WACV, 2026.
[4] C. Edge et al., “A Generative Approach for Detection-driven Underwater Image Enhancement,” arXiv:2012.05990, 2020. (DUnIE)
[5] R. Mukherjee et al., “The Common Objects Underwater (COU) Dataset for Robust Underwater Object Detection,” IROS, 2025.
[6] K. de Langis, M. Fulton, and J. Sattar, “Video Diver Dataset (VDD-C) 100,000 annotated images of divers underwater,” DRUM, 2021. doi:10.13020/6qrp-wy09.